

JAKUB KIERZNOWSKI

Junior Software Engineer | Software Developer Intern at IBM
CS & Intelligent Systems Student at AGH

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SUMMARY

Software engineer who builds products and the AI inside them. At IBM I optimise graph and tree traversal in Data Lineage over enterprise metadata; after hours I build and support InstalDesk, a live B2B SaaS with a native integration to the Polish KSeF e-invoicing system. On the AI side: an LLM agent with Cypher retrieval tools over a 47,031-node biomedical graph, reinforcement-learning agents negotiating in a power-grid simulation (paper in preparation), and an engineering thesis on deep learning for 2D-image to 3D-mesh conversion. **I am looking for a junior software engineer role where product engineering meets applied AI.** Final-year Computer Science and Intelligent Systems at AGH, GPA 4.74/5.00, graduating January 2027.

Java 21 · Spring Boot · Spring Security · PostgreSQL · Neo4j · Python · FastAPI · Docker · RabbitMQ and MQTT · Jenkins · GitHub Actions

EXPERIENCE

Software Developer Intern

IBM Software Lab · Kraków · hybrid
Feb 2026 – Present

- Java and Spring on Data Lineage: optimise performance over large enterprise metadata graphs by implementing graph and tree traversal algorithms. (Client data and exact figures are under NDA.)
- Cut the pipeline test run from 20 minutes to 4 by making the automated pytest suite asynchronous — parametrised cases, session-scoped fixtures.
- Own the Jenkins job scheduling for the pipeline.
- Daily Scrum in an English-speaking cross-functional team.

IT Team Lead, Web & Workflow Automation

Student Council · WRSS WEAlIB, AGH
Mar 2025 – Present

Council member since Mar 2024, leading its IT team since Mar 2025. I maintain the council website (Vue, TailwindCSS) and the VPS it runs on, and built internal automations on self-hosted n8n against Slack and Google APIs over OAuth 2.0. Event coordinator for the VOLT 2025 tech conference, eight partner companies including Aptiv, PwC Polska and Cytiva, and part of the organising team for VOLT 2026.

Team Mentor and Community Leader

Zwolnieni z Teorii · remote
Sep 2022 – May 2025

Mentored 35+ student teams per edition, from planning to the final presentation of their social-impact projects. Later ran a participant community (ZWzT × PwC certificate) and supported olympiad entrants.

Game Developer

AGH Code Industry · Dec 2023 – Jun 2024

Gameplay mechanics in Unity and C# for “Student Trainer”, a game built by a student team under a mentor.

PROJECTS

Code at [<repo>](https://github.com/qualv13)

InstalDesk, B2B SaaS for Polish installation contractors

instaldesk.pl · live since Apr 2026, v1.9.0, pilot users · closed source · Python, FastAPI, PostgreSQL, SQLite, Stripe, KSeF API

- Native KSeF integration: API v2 against the FA(3) schema, KSeF-token auth on web, XAdES-BES certificate auth on desktop — filed straight to the Ministry of Finance, without a third-party integrator.
- A web PWA and a Windows desktop client share one account. PostgreSQL in the cloud, SQLite locally so the desktop client keeps working offline.
- Per-tenant isolation, KSeF tokens encrypted at rest. Stripe billing over card, BLIK and Przelewy24, a 14-day trial without a card, renewal flows, transactional email, PDF generation, XLSX and CSV import and export.
- Built and supported solo, including user contact.

DrugPath, biomedical knowledge-graph agent

neo4j-agent · Neo4j Aura Agent Hackathon 2026, top 6 · Neo4j Aura, Cypher, Python, LLM agents

An LLM agent with four retrieval tools over the Hetionet biomedical graph (47,031 nodes, 293k relationships): three Cypher queries and a similarity search, driven by a system prompt, walking drug → gene → pathway → disease paths to explain how a drug works and where it might be repurposed. Educational project on public research data. Demo at qualv13.github.io/neo4j-agent.

IoT fleet backend

IoTServerApp · Java 17, Spring Boot 3.3, PostgreSQL, RabbitMQ/MQTT, Protocol Buffers, Docker

Layered Spring backend for managing smart-lamp fleets. JWT auth with refresh tokens, MQTT command and telemetry topics over RabbitMQ, a Protocol Buffers device protocol, OTA firmware delivery from S3, scheduled device jobs. Full Docker Compose stack.

Multi-agent reinforcement learning for power-grid negotiation

research project · AGH, paper in preparation · Unity, C#, PyTorch, PPO, ONNX

A simulation environment where agents negotiate power distribution across a grid under load. Agents trained with Proximal Policy Optimization in PyTorch and served back into the Unity simulation through ONNX. Two-person project: I gathered the data, built the environment and ran the training.

Hardware Hub, AI-native inventory and rental system

hardware-hub · recruitment assignment, live · Python, FastAPI, Neo4j, Vue 3, Gemini, Docker

Compatibility is derived by traversal (device → shared port → machine → OS) rather than stored, so a search returns only hardware that actually connects, each hit with the path behind it. Gemini tool calling for semantic search and an inventory auditor, with deterministic fallbacks. A golden query set scores the graph engine against the flat one; 72 pytest tests on CI.

BPM-first workout playlist generator

fit2beat.toadres.pl · HackYeah 2025, top 8 of 65+ · Python, Flask, Docker

Builds playlists from a target BPM rather than from genre. Nobody else in the biohacking track approached it that way. Team of four, 24 hours, 3.9/5 from the jury.

Graph and deep neural networks

GGSN · Graph and Deep Neural Networks, AGH coursework · PyTorch, TensorFlow, scikit-learn

Graph neural networks (GCN, GAT, GraphSAGE, GIN) over the Yelp dataset, benchmarked against one another. Transfer learning with Grad-CAM, DistilBERT sentiment classification, and network compression across seven architectures.

SKILLS

Tech

Core: Java 21, Spring Boot, Spring Security, Hibernate/JPA, Maven, JUnit 5, PostgreSQL, SQL, REST APIs, OOP and design patterns, algorithms and data structures (graph and tree traversal, sliding window, complexity analysis).

AI: LLM agents and tool calling, retrieval over knowledge graphs (Neo4j, Cypher), prompt engineering, self-hosted models (Ollama), PyTorch, reinforcement learning (PPO), graph neural networks, TensorFlow, scikit-learn.

Working knowledge: Python (FastAPI, pytest), Docker and Compose, Linux/Bash, Git, Jenkins and GitHub Actions CI/CD, RabbitMQ, MQTT, Protocol Buffers, Testcontainers, OAuth 2.0, AWS S3.

Also used: Vue with TailwindCSS, C# (Unity), C++, n8n.

Soft

Agile and Kanban on GitHub project boards at IBM, Jira before that; daily Scrum in a cross-functional English-speaking team. Lead event coordinator for VOLT 2025, including the partnership deals with eight tech companies.

Languages: Polish native, English C1 (working language at IBM), French A2.

CERTIFICATIONS

Neo4j Certified Professional and Building Agents in Neo4j Aura, both 2026.

Amazon Junior Software Developer (Coursera specialisation): Data Structures and Algorithms, Programming with Java, Application Development.

INTERESTS

Snowboarding, sailing, a self-hosted homelab running Docker and quantised local models, game dev, board games.

EDUCATION

Computer Science and Intelligent Systems, BEng

AGH University of Krakow · Kraków
Oct 2023 – Jan 2027

GPA 4.74/5.00. Year Representative since the 1st semester.

Extended Mathematics, Physics and Computer Science

V High School · Kraków
Sep 2019 – Jun 2023

C++, competitive-programming problem solving.